

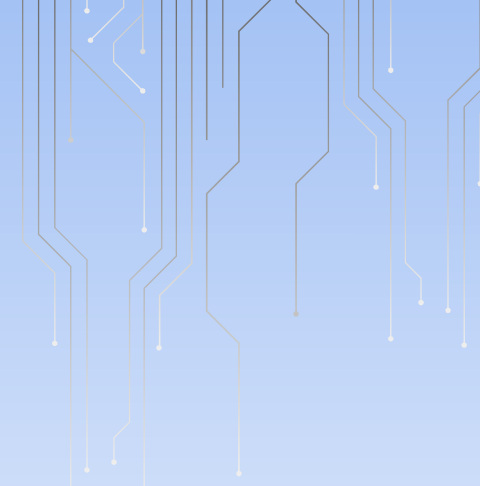
SDR-Based DDR Air-Gap Data Exfiltration

Exploiting Electromagnetic Side-Channels for Covert Communication



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Air-gapped networks



What are they?

Systems physically **isolated** from unsecured networks such as the internet. They are reserved for **high-security** environments and the assumption is that since they are physically disconnected from other networks they are **virtually immune** to remote attacks.

The reality

All electronic devices inherently emit electromagnetic radiation during operation and these emissions can act as a **side-channel**. This project's goal is to show that data exfiltration is **possible** with a simple user-space program and no special equipment.





The leaking mechanism

Maxwell's Equations

A time-varying electric current inevitably generates a magnetic field. This means that rapid change in current will produce electromagnetic signals.

DDR memory

Modern DDR memory buses operate at extremely high speeds, generating significant electromagnetic energy that translate to signals.

Memory as an antenna

By effectively "bit-banging" the memory bus with specific software patterns, we can intentionally shape these emissions into a radio signal.

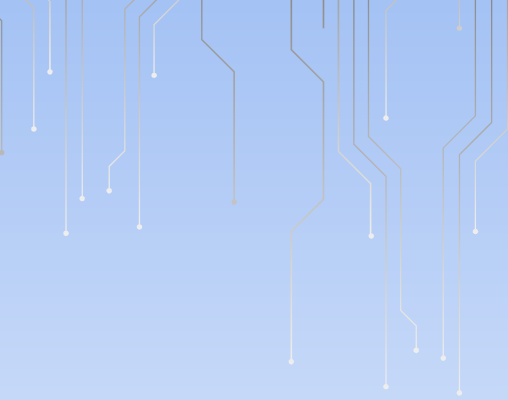
Setup

Victim Environment

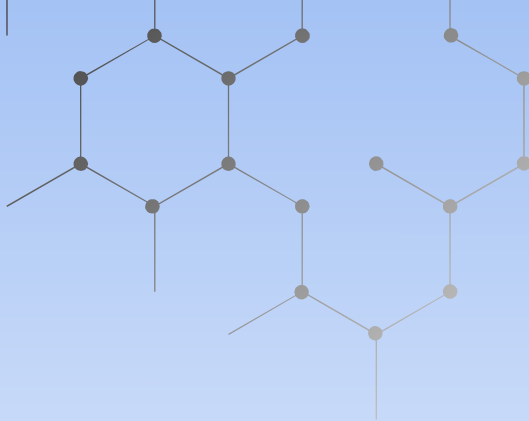
The target is a standard air-gapped workstation where our unprivileged malware exploits the physical memory bus to broadcast signals through the noise of legitimate background processes. This attack requires no hardware modifications or special permissions.

The Channel

We bridge the air gap by modulating memory bus emanations to reach a passive receiver about 1 meter away. Because the receiver emits no radio frequency energy, the interception is completely undetectable by standard counter-surveillance measures.



Challenges



Background noise

Capturing device noise

Distance

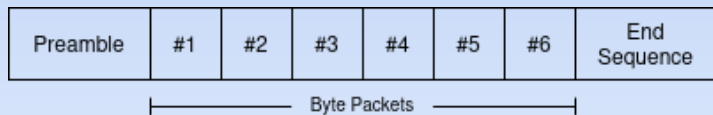


Modulation

```
static inline void steroids() {  
    __m128i* buffer_ptr = (__m128i*) buffer;  
  
    for (uint32_t i = 0 ; i < BUFFER_SIZE ; i++) {  
        _mm_stream_si128(  
            &reg,  
            *buffer_ptr  
        );  
        buffer_ptr = buffer_ptr + 1;  
    }  
}
```

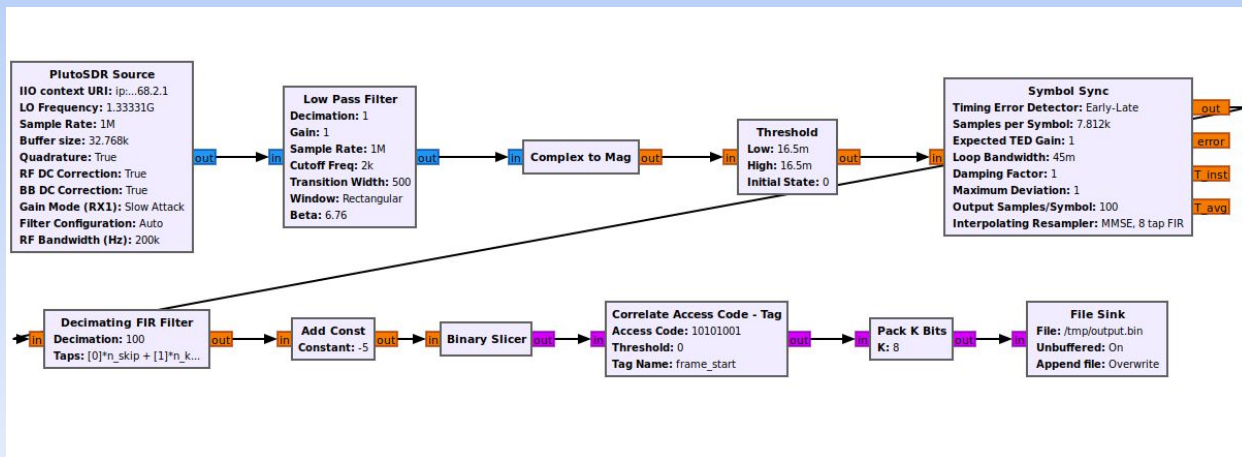
- Modern CPUs effectively silence the memory bus by trapping data in internal **caches**, preventing the generation of detectable electromagnetic signals.
- We overcome this by using the `_mm_stream_si128` intrinsic to execute **non-temporal** stores, which instructs the CPU to bypass the cache hierarchy entirely.
- This forces every write operation directly to the **physical DRAM**, generating the distinct electromagnetic spikes necessary for our receiver to detect the transmission.

Protocol structure

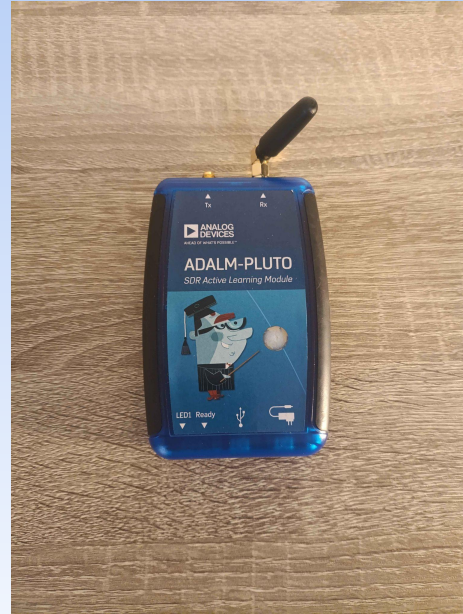
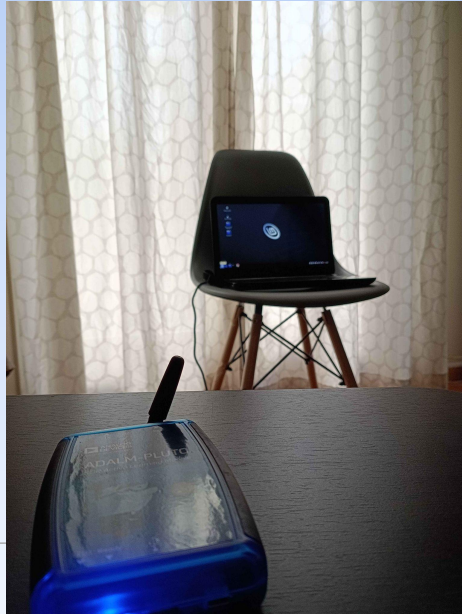


- **Preamble:** Synchronizes the receiver to the correct bit alignment before the payload begins.
- **Payload:** Arbitrary binary data transmitted sequentially, bit-by-bit (MSB or LSB first depending on implementation).
- **End of Message (EOM):** Signals the receiver to stop recording and flush the buffer.

Demodulation



Experimental environment



Transfer times for 128-bit key

Bitrate (bps)	Duration	Delay
4 bps	32	250ms
8 bps	16	125ms
16 bps	8	62.5ms
32 bps	4	31.25ms
64 bps	2	15.625ms
128 bps	1	7.8125ms

Bit Error Rate (BER) %

Bitrate (bps)	25cm	50cm	100cm
4 bps	0%	0%	0%
8 bps	0%	0%	0%
16 bps	0%	0%	0%
32 bps	0%	0%	0%
64 bps	9.67%	16.4%	29.68%
128 bps	51.2%	failed	failed

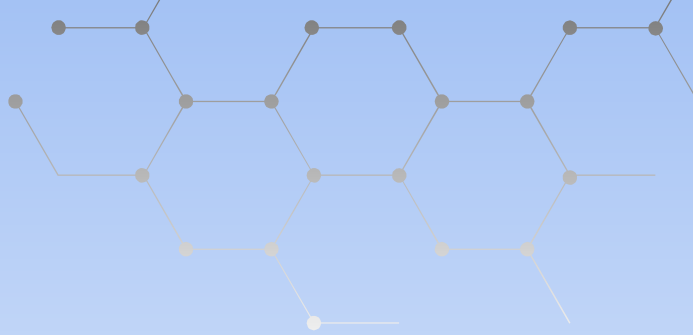
Conclusion and Future work

Proof of concept

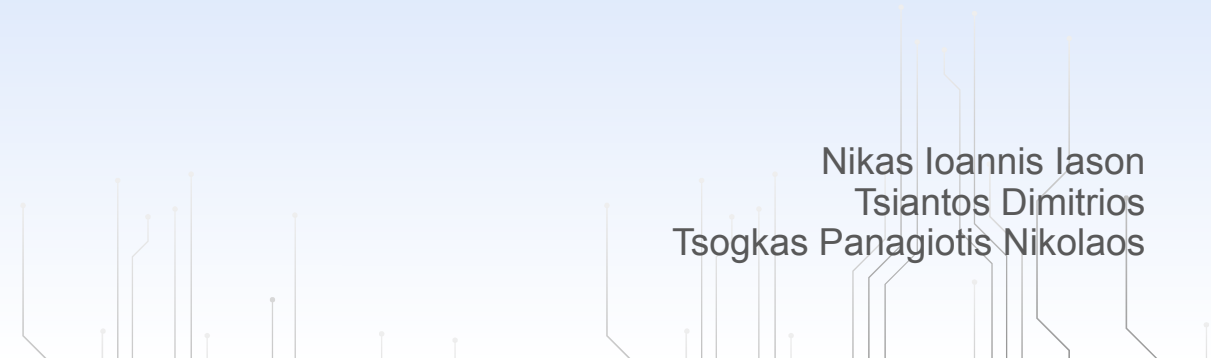
Our results validate that physical isolation is not a guarantee of security. The physical layer remains a viable attack vector even when the network layer is removed.

Improvements

To improve protocol robustness, we propose using a fixed-length header, ensuring that payloads containing the termination byte do not stop the transmission. Furthermore, we recommend implementing error correction methods to achieve better accuracy.



Thank you!



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